ments. Many of the environmental requirements, like the conservation of energy and other resources, also directly impact profitability. Even in cases where such direct relation is not present, consumers prefer a product that is less harmful to the planet.

Analysis of the profitability often poses a problem as different expenses happen at different times and may not be apparent right at the beginning. One needs to account for all the revenues and expenses to get an idea of the profitability.

It is easy to get an idea of the material that goes directly into the product, which is normally available as bill of material during the design process. We can estimate labour requirement by comparing it with similar processes.

There are many published time standards, like the MTM time standards for manual processes, or the Boothroyd and Dewhurst Assembly Time Estimation for automated lines, that help us to estimate these numbers.

From our estimation of machine and labour capacity, we get the number of machines and labour required for our target production volume. Here, the process planners must take make-or-buy decisions.

For manufacturing our designed product, we need to purchase or lease machines and equipment. In addition to processing equipment, we also need equipment for storing and handling raw materials, work-in-process materials, and finished goods. We also need land for all these materials and equipment.

Money comes with a cost. When we keep some money in the bank, we earn interest. Similarly, when we take a loan, we need to pay interest. In business, the entrepreneur must invest in all the land, machines, and materials. He earns his revenue after making the items, marketing them, and receiving payment from the dealers. There is a gap between the entrepreneur investing his money and earning the revenue.

Business needs working capital to run its operation and this money comes with its cost of capital. We need to estimate the amount of money needed as well as the time of its requirement, and when we get back the money as revenue.

TABLE 1
NPV CALCULATION

Year	Description	Amount (₹)	NPV (₹)
0	Purchase of land and building	-1,500,000	-1,500,000
1	Purchase of machines and equipment	-1,000,000	-909,091
1	Working capital	-3,000,000	-2,727,273
1.08	Earning (every month)	1,000,000	
	Eq. one-time earning	12,048,193	10,866,599
	Net NPV		5,730,235

Profitability performance indicators

There are a few standard performance indicators that help us to understand the economic viability of our product idea and compare it with other businesses. Let us discuss a few popular indicators and compute to see how our product idea compares.

Net present value. Net present value (NPV) starts with a presumed cost of capital. The investor has the option of keeping his money in a bank, or some bonds, and earning interest. He needs to earn something better from the business than the return he will get from his safe investment in a bank's fixed deposit or bonds to justify the effort and risk he takes with his money.

The notional minimum acceptable interest rate that the entrepreneur will accept is the cost of capital. We estimate the time and amount of various investments and earnings and convert those into the monetary value at the present date with interest charges using the cost of capital. The total of all these present values gives us the net present value.

Mathematically,

$NPV = Sum\ [R_t/(1+r)^t] - Sum\ [I_t/(1+r)^t]$

where, R_t is the revenue earned in period t, I_t is the investment made in period t, r is the cost of capital.

In the initial phase, we start our analysis with estimates and get an idea of the economic viability of our idea. Let us take the case of our charge controller. As a rough estimate, the raw material of the product will cost around ₹600.

We need to invest approximately ₹1,000,000 in machines and equipment. Investment in land and building will be around ₹1,500,000 Lakhs. We estimate that factory building will take one year time.

We hope to sell around 100 pieces per day at ₹1,000, excluding taxes and commissions. We estimate our business cycle from raw material procurement to getting the money from the dealer will take approximately 30 days. This gives our working capital requirement of ₹3,000,000, as calculated below.

30 days × 100 pcs/day × Rs1,000/pc = ₹3,000,000

Let us say we expect to make a profit of ₹400/pc. Assuming 25 days working in a month, our investment will generate an income of ₹400/pc × 100 pcs × 25 days = ₹1,000,000 every month. For ease of calculation, we convert all recurring earnings or expenditures into a one-time amount by dividing it by the interest rate.

Details of NPV calculation are given in Table 1. If we assume our cost of capital to be 10% per year, we find that our project has a robust NPV of over ₹5.7 million. Hence, we can proceed with this product with confidence.

Return on investment (RoI). In our previous NPV calculation, we assumed a cost of capital. RoI calcula-